

Final Paper

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1 Prompt

This paper is a chance for each of you to explore computer science in a more educated environment. The paper is such that you should find an interest that you personally possess and write about that. It is often the case that finding something you are interested in and realizing it is difficult. In particular this paper tends to stretch peoples ability to self reflect on what is truly interesting to them. For me in particular I hold many interests, chiefly among them at the moment is topological type theory. This is a field of mathematics that allows for algebraic, general solutions over large sets of continuous spaces such as $\mathbb{R}$. It is obvious to me that I hold interest in this topic given that I am a computer scientist and hold particular interest in set theoretic problem spaces. For a student this may look a bit different, look into hobbies or personal subjects that hold your interest. It is more difficult to find a subject not connected to computer science in some way in the modern world.

2 Rules

As these papers must be gradable and solvable. The paper format is roughly %80 written English portion. This is where you will explore your interest and explain either some intuition for the subject or explore its boundaries. The remaining %20 of the paper will be a logical explanation using Python in some way. You have learned about Python this semester and how to use it for multiple mathematical theorems, make an attempt to explore your

chosen subject field using Python. An example would be as some kind of socket programming library for a subject such as networking. Something like exploring the `import(socket)`¹ library and its various use cases with some examples.

¹note that if your subject is networking this is a good idea